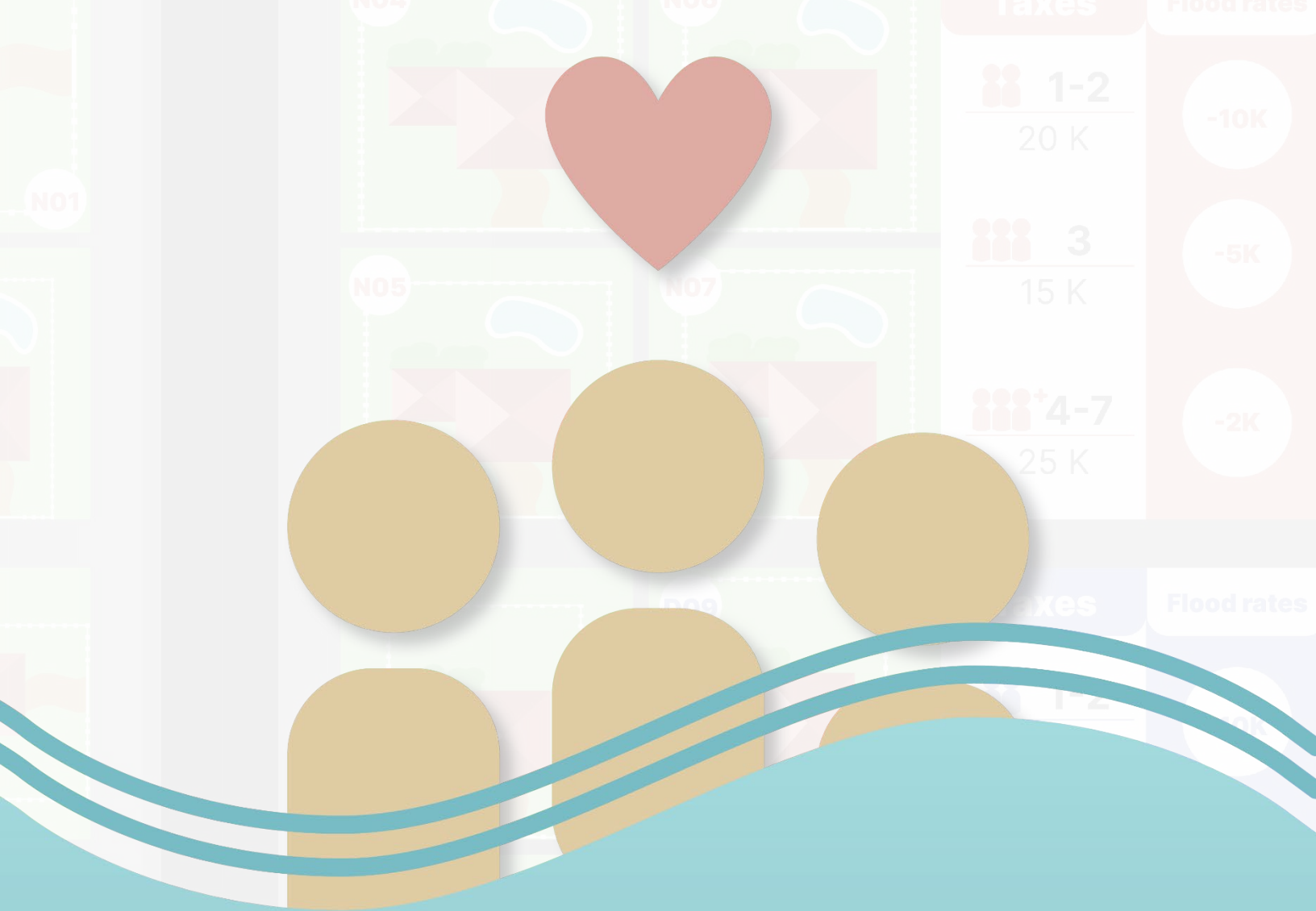




Where We Move

Facilitator Manual

TU Delft Game Lab

February 2026



Where We Move is a housing data collection game that supports stakeholder organisations and residents in joining efforts for climate action by exploring how homeowners' perceptions and resources shape player's adaptation choices. The game aims to achieve the highest score possible by increasing satisfaction throughout the game or earning points for the available income left at the end of the game.

Don't explain to players unless they ask: Living costs and house-rating expectations are somewhat proportional with a player's income. High-income players face higher living costs and expect a higher house rating, representing the expected quality of their house in the game. The game includes affordable houses for low-income players.

Learning Goals

Players take the role of a homeowner to:

- ❑ Learn about flood risk and how to protect your (game) house
- ❑ Reflect on how risk perception and game resources shape players' choices
- ❑ Explore ways to empower residents such as the game homeowners to take more adaptive actions.

Expected Gameplay Benefits

The game focuses on the following actors, each with their own expected benefit from the gameplay:



**Residents
representatives**

Make residents (also students), and representatives of stakeholder organisations take the homeowners role to be aware of their available resources to take action (i.e. knowledge about the measures, welfare type, house location, lack of flood experience, etc).



Teacher

Enable students (future homeowners) and professionals to reflect on the homeowner's role and capacities to join efforts for flood protection with stakeholder organisations.



**Stakeholder
representatives**

Provide stakeholder organisations such as municipalities and insurance associations with a safe environment to explore possible policies for strengthening residents' action capacities.



Researcher

Gather per-game round data on the following questions to give insights for flood risk communication to the above actors, for example:

- ❑ How did players spend their spendable income on average?
- ❑ In which game area do players want to live mostly?
- ❑ Which measures did players choose at every round?
- ❑ How did welfare type affect their choices?
- ❑ Did the player's risk profile have any influence on their choices?
- ❑ Did the game flood experience influence the player's choices?
- ❑ What strategies did players use in the game? Do players focus on improving their bought house for personal satisfaction, making flood adaptations, or moving to another house?

Facilitator Manual Overview

p₇



Session set-up

- ☒ To accordingly prepare for the game session, ask to the game coordinator about the session set up including:
 - ☒ Ethical and privacy agreements,
 - ☒ Table you were assigned to,
 - ☒ Number of rounds to play,
 - ☒ Whether moving is allowed or not every round
 - ☒ Game scenarios to play, if any and
 - ☒ Number of players per table.
- ☒ Get familiar with the [facilitator website](#) and the [player's website](#) before the gameplay. Remind the game coordinator to clear any test game records before the session to start data collection.

p₉

Table set-up



- ☒ Arrange the room with the number of tables to play, with 6 to 8 seat per table.
- ☒ Check that all game elements per table are complete:
 - ☒ 8 player labels
 - ☒ 9 improvements cards x 8 players.
 - ☒ 8 pairs of pawns x 8 different colours.
 - ☒ 1 board addition for the news x flood reduction for all areas.
 - ☒ 1 board addition for the news x changes in game taxes.
 - ☒ 2 x 6-sided dice and 1 x 10-sided dice, where 0 means 10.
 - ☒ 1 coin x 8 players x 6 improvement cards providing flood protection.
 - ☒ 4 coins x 8 players x 3 satisfaction dedicated improvement cards.
- ☒ Put the improvement cards next to the player login and pawns organised per category: only satisfaction measures, measures to add capacity to drain rainwater, and measures to prevent the water entering/ damaging your house
- ☒ Access the [facilitator website](#). Login to the session agreed upon. Username: fac# and password:fac#, where # is the table # to facilitate.

p₉

Player set-up



- ☒ Access to the [player's website](#) for one playerID (e.g., t#px) to check that the credentials work.
- ☒ Arrange each player's seat with a label, two pawns and the measures cards.
- ☒ Put the coins on top of the icon of every improvement card.
- ☒ Ensure every player has a different pawn colour pair. Every player gets randomly assigned a game income (i.e. welfare type) when sitting.
- ☒ There are, in total, 6 types of welfare to play assigned by the game coordinator. When 8 players, the middle welfare repeats in two players.

p₁₀

Facilitator set-up

- ☒ Arrange on a separate/ side table for the facilitator:
 - ☒ Laptop with the facilitator website.
 - ☒ Houses per community and house address.
 - ☒ Board additions for the news.
 - ☒ Papers for the bid cards.
 - ☒ Debriefing material.
- ☒ Set up your phone as back-up for the player's website, in case any of the player's phones give problems.

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Gameplay

Welcome and login

- ❑ Welcome every player when sitting, introduce yourself and the player's game role by explaining what the credentials and the pawns are for.
- ❑ After the plenary game introduction, check that all players in your table complete their pre-survey.
- ❑ Log in to the facilitator side and check in **PLAYER OVERVIEW** that all players login.
- ❑ Go to **NEWS OVERVIEW** and read **Round 0 News** with the goal of the game and explain the scores. Your news text is longer. In '*' what the players read.
- ❑ Clarify any question that players may have after the game introduction and news.

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1. Rounds

- ❑ Click the facilitator's **START NEW ROUND**.
- ❑ Ask players to click **START ROUND** and review their budget and expectations.
- ❑ Explain the values that are **the same every round** in their:
 - ❑ Budget (round income and living costs) and
 - ❑ House expectations (house rating and maximum mortgage).

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2. News

- ❑ Click the facilitator's **ANNOUNCE NEWS** but only read the news after players click **READ NEWS** and have 30 secs to read by themselves.
- ❑ Ask players what they remembered from the news and clarify as necessary.

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3. Show houses

Open house market

- ❑ Click the facilitator's **SHOW HOUSES** and explain the house addresses' N#, D#, and U# community naming. Players click on the **VIEW HOUSES** to choose the desired houses from their drop-down menu.
- ❑ Remind players what is the desired house rating and maximum mortgage, so players can choose a house.
- ❑ Players can increase their satisfaction by buying improvements later in the round.
- ❑ There is no need to target the ideal house rating initially; moving is possible again in the last round when new houses become available. Moving costs 1 satisfaction point.
- ❑ Ask players to keep the savings to pay taxes (about 20k), mortgage (10% of the value of the house), living costs (varies per role), buy improvements and damage.
- ❑ If players go into debt, players lose a satisfaction point at the end of the round.
- ❑ Give everyone 1 minute to choose their house and put the marker on the board when you ask them to do so. If players want, they can check the house card before buying.
- ❑ If two players choose the same house, there is a bid whose value is only visible to the facilitator. The player with the higher bid wins.
- ❑ Only if there is no other house, allow a player to go into debt to buy a house.

p15

3. Allow buying

House check

- ❑ After all players agree on the house to buy and have the pawn on the board, click on the facilitator's **ALLOW BUYING**.
- ❑ Player's click **BUY HOUSE** to send a buying request. When receiving the request, check if the price is the same as the house card.
- ❑ The house request shows a red warning when there is a difference with the initial price set in the administrator website.
- ❑ Check if a bid or discount applies, so you can comment before approval or deny if necessary.
- ❑ When bidding, check that the player updated the house value with the bid.
- ❑ If not yet, give the house card to the player.

p15

3. Finish buying

Players move in

- ❑ **FINISH BUYING** only when you have approved the request of all active players. Late requests are possible but may produce game errors.
- ❑ Ask all players to click **MOVE IN** or **GO BACK** if you approve or reject their request.
- ❑ If moved in, make players notice the change in their spendable income.
- ❑ Give players their game house and ask them to put it next to their player's label so that it is visible to all other players in the table.

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3. Allow selling

Players stay in or move out

- ❑ Selling is possible from round 2 onwards but only encouraged in the last round.
- ❑ In round 2, the facilitator will only approve moving when a player starts badly on a debt and there is a house.
- ❑ Players choose first the house they would like to move before consider selling their house.
- ❑ Ask players to scroll down all the way to the end if they simply want to **STAY IN**.
- ❑ Whether players **STAY IN** or **MOVE OUT** of their house, follow the abovementioned buying steps. Players can buy each other's house if they agree; otherwise, the bank sell it at the original price.

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3. Calculate

Players pay taxes

- ❑ Click the facilitator's **CALCULATE TAXES** and ask players to compare with their board positions.
- ❑ Ask players to check what they paid by comparing the taxes to pay according to their house choice and number of players that moved into the same community.

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4. Allow improvements

- ❑ Players can choose between buying house improvements to avoid water to enter or increase rain drainage, increase their house or personal satisfaction with house or personal improvements and flood insurance.
- ❑ Ask them to put the icons of what they bought on top of the house card.
- ❑ Satisfaction-only measures can be bought many rounds. Sand bags remain active until used.
- ❑ Let players choose as freely as possible without prompting them in any direction but make them aware of all possible options. Players cannot go into debt to buy improvements.

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5. Player perceptions

- Click the facilitator's **SHOW SURVEY** and read the question and answer aloud so players do not choose randomly. Players can continue only if choosing an option.

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6. Floods and Damage

- One player rolls the six-sided dice for the river volume, and another player rolls the ten-sided dice for the rain volume.
- Ask players to subtract from the volume/flood level they rolled, the flood protection in their community plus the protection their improvements give.

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7. Summary

- Player's **WAIT FOR THE DICE** button allows them to check if any damage.
- If damaged, for rivers players pay xk per level above and get as much satisfaction as a penalty. For rain, players pay one-point damage regardless of the level above. If flooded due to river, an extra satisfaction penalty applies because of the damages in the community.
- Go to step 1 to click on the facilitator **START NEW ROUND** until round 4 ends.
- At the end of round 4, the remaining money adds to the satisfaction score.
- News for every round are as follows in short:
 - Round 2: Flood damages explanation reminder
 - Round 3: Climate change.
 - Round 4: Taxes Naturcity +5k (add taxes for Naturcity)

p29

Game settings

- If necessary, check the game settings for the:
 - Game news text in English and Dutch
 - House market characteristics

p34

Game credits

- If necessary, check the game settings for the:
 - Game news text in English and Dutch
 - House market characteristics

Game Session Preparation

Game session parts, facilitator, and player tasks

Every session lasts maximum 3 hours, depending on the agreements made, and includes roughly the following parts. Gameplay rounds and survey questions were discussed beforehand to focus on the interests of the relevant organisations. After the participants are divided into tables of 6 to 8 players, each with a facilitator the game session follows the following parts:

		Facilitator Tasks	Player Tasks
Introduction 5 min		Game coordinator's welcome and ice-breaker questions	Where do you live? Do you own or rent? Any experience with serious gaming?
Player consent for data collection 5 min		Slides with Why WhereWeMove introduction	Before starting the game, players scan the QR code.
Pre-survey 10 min		Pre-survey about your flood risk perception	Players use their player ID to complete the presurvey.
Gameplay 90 min		The facilitator of each table: - Oversees the gameplay on the website to support players when necessary. - Announces if a climate event occurred to check if any resident is affected. - Acts as the bank for buying and selling.	During the game rounds, players will play by tracking their choices on the game website.
BREAK 10 min		Enjoy your beverage!	

Post-survey

10 min



The facilitator announces the winner and invites participants to reflect on their game experience through the post survey.

Players fill out the post-survey

Table debriefing

30 min



Facilitator takes notes in the computer and provides reflections sheets to one of the players to take notes from the discussion.

❏ Player with the highest and lowest score share their experience. Other players add as necessary their limitations and motivations to act in the game. **(10 min)**

❏ Role play taking the role of an actor to address limitations and motivations to act in real-life **(10 min)**

❏ Players make a strategy to start joining efforts within the next five years. **(10 min)**

Plenary debriefing

15 min



The game coordinator asks players to share insights from discussions.

Players present at each other's tables, their debriefing highlights.

Game elements

Before introducing the players to your table, you must know what game you will play.

Learning Goals

Players take the role of a homeowner to:

- ❏ Learn about flood risk and how to protect your (game) house
- ❏ Reflect on your risk perception and game information to make choices
- ❏ Explore ways to empower residents to take action.

Game scenarios

Currently, only one scenario with ALL INFO is available. In earlier versions, we offered two:

- ❏ ALL INFO: Information regarding the flood protection of available improvements was not given.
- ❏ LIMITED INFO: Only a description of each measure, without its protection level.

Other scenarios can be enabled on request—such as reduced measure costs or added options like flood insurance. Future versions may also include gameplay with or without a flood-risk discount on house values.

Table setup

The game box contains the following elements: check if present and stored as such after the game play:

Per Facilitator:

- 1x debriefing package per number of tables to facilitate.

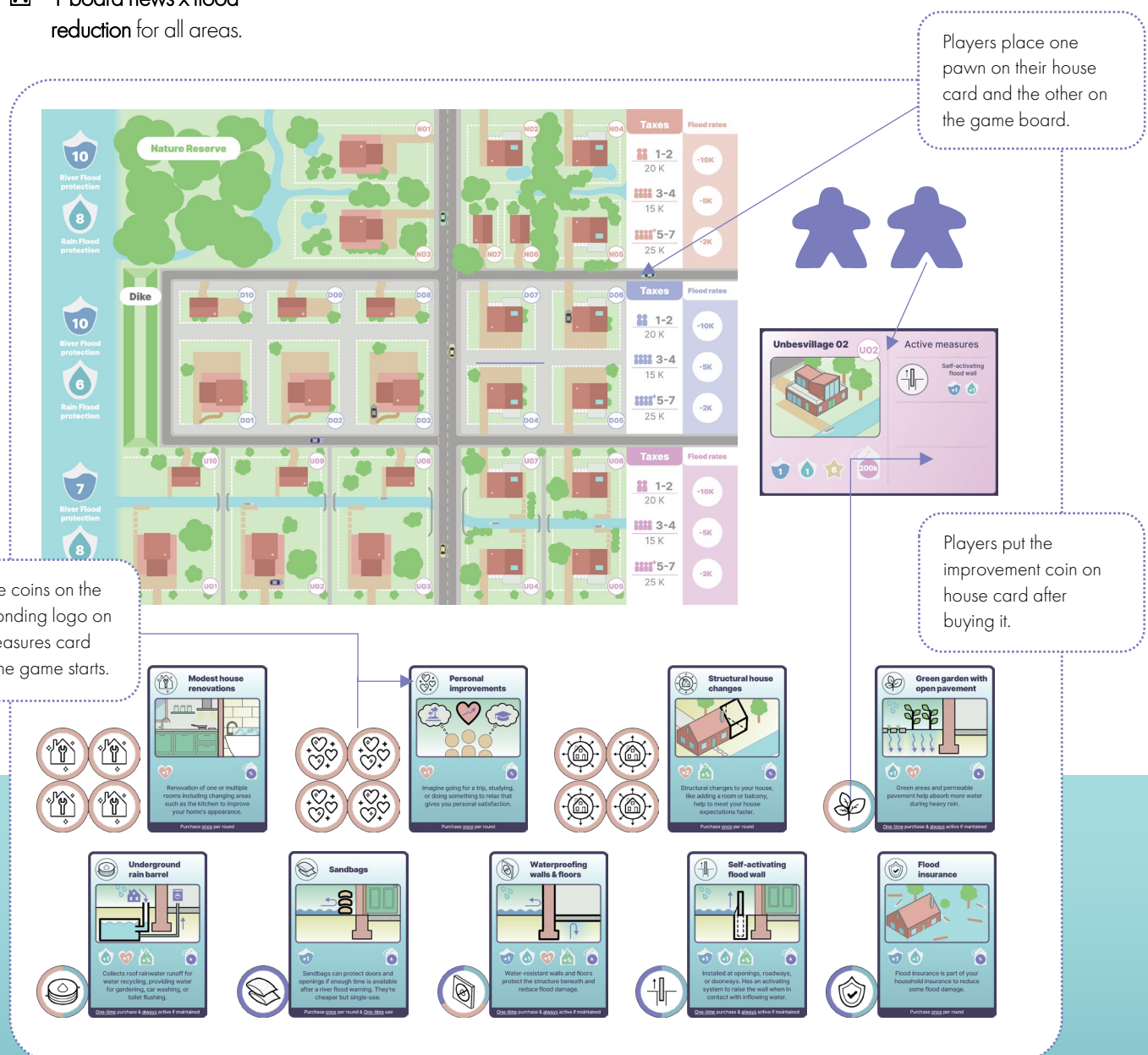
Per table:

- 1x game board
- 2x 6-sided dice for river flooding
- 1x 10-sided dice for rain flooding, where 0 means 10
- 1 board news x flood reduction for all areas.

- 1 board news x taxes for Naturcity.
- 3x cubes for house discounts (to use if game setting is active)
- 7x Naturcity house cards
- 10x Diketown house cards
- 10x Unbesvillage house cards
- 16x house bid cards

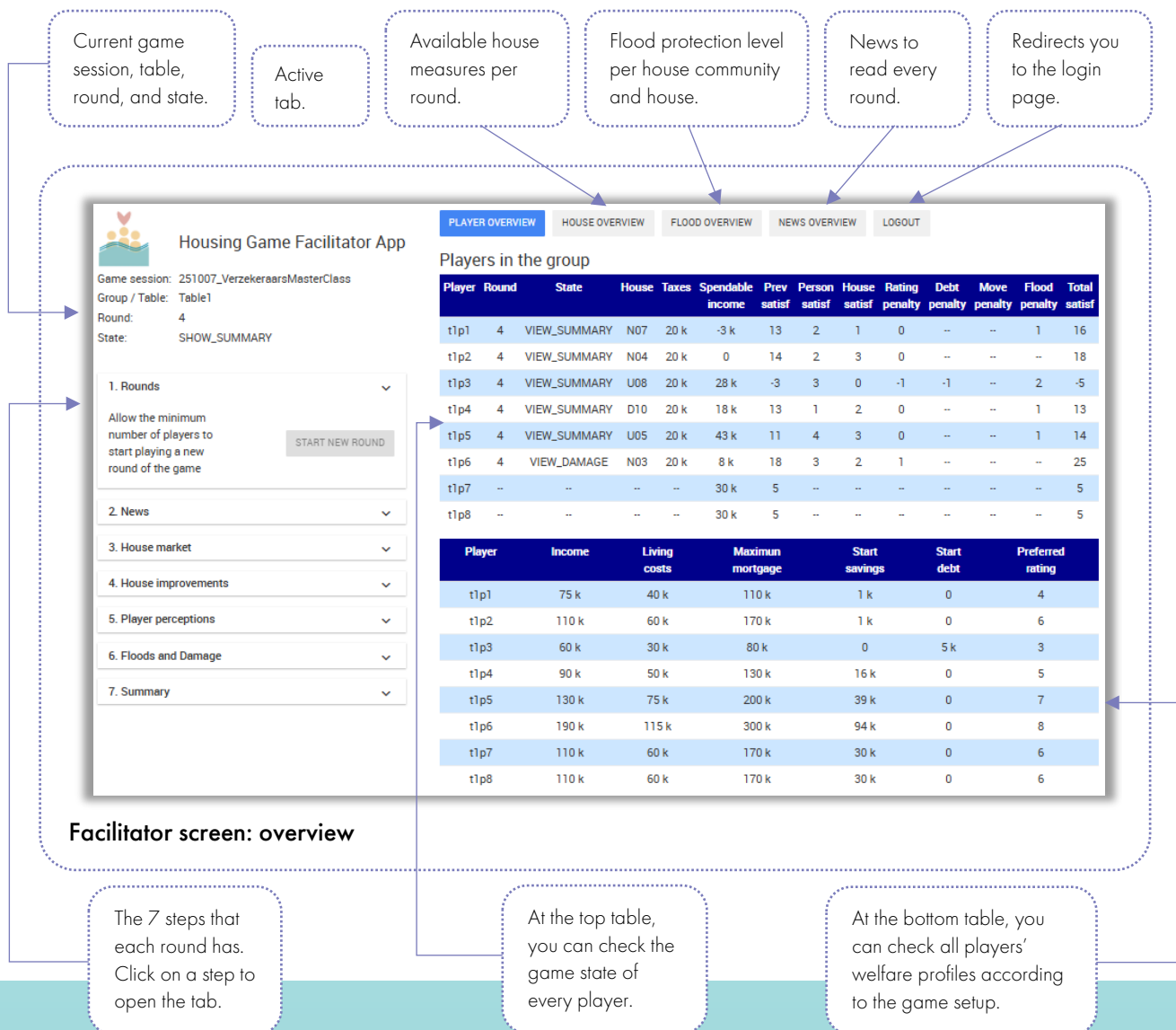
Per 8 players:

- 1x PlayerIDs with login details
- 2x pawns in one color for every player (8 players)
- 9 improvement cards
- 1 coin x 6 improvement cards providing flood protection
- 4 coins x 3 satisfaction dedicated improvement cards



Facilitator's website

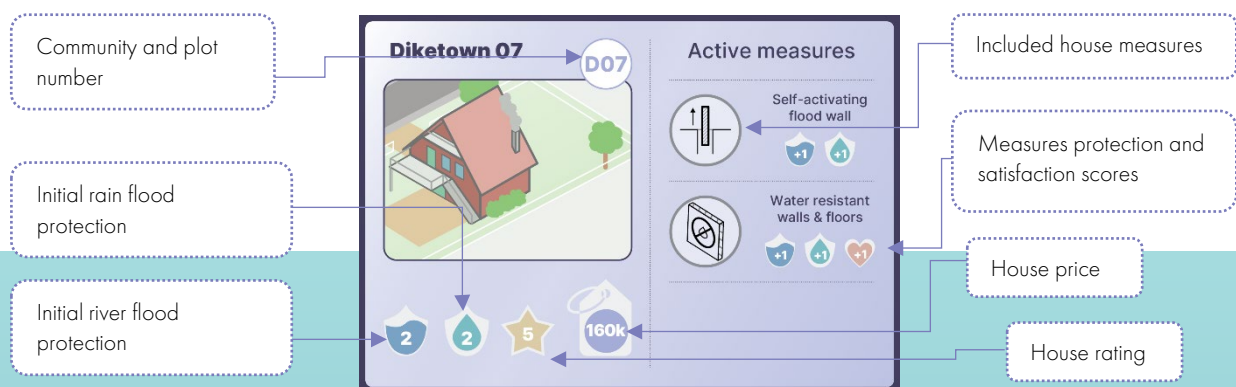
After logging into the facilitator website, the top left of the screen shows the current game session, table, round, and state. The left accordions show the 7 steps every round has. Steps range from starting a new round to showing a summary. Various overviews are given in the tabs, where the blue tab represents the tab that is currently being viewed.



House market setup

Keep in mind the following aspects when announcing the game market:

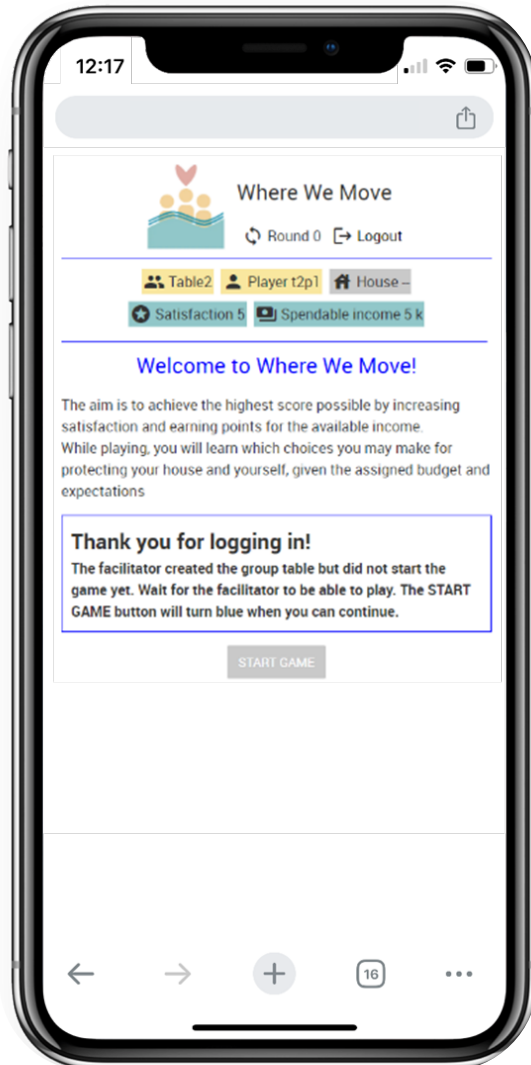
- ❏ Keep the house cards on a pile per community and ordered per house ID to quickly exchange them with the players.
- ❏ The houses available per round change as set up by the game coordinator. Not all houses are available at once because some are "occupied".
- ❏ All houses in round 1 start without available measures, hence the additional house protection is zero. This means that the player is only protected by the board flood protection, according to the house location.
- ❏ If buying a house with lower rating, there is 1 point satisfaction penalty only when buying. There is no need to target the ideal house rating initially; Players can increase their satisfaction points by buying measures later in the game.
- ❏ Suggest the players to keep 50k in savings to pay the mortgage (~10k for 10% of the house value chosen), taxes (around 20k), improvements (~20k)
- ❏ Houses that are not sold will not be available again unless indicated otherwise by game coordinator.
- ❏ Moving costs 1 satisfaction point.
- ❏ In the last round, the house market is updated with houses that include measures. The total flood protection is the protection from the game board community and the included measures combined.



Playing the Game

Welcome to the game table

Read the Round 0 News “Welcome to Where we Move – Table X, this is a serious game involving houses ...” and make sure that all players have log in to click the START NEW ROUND button in the facilitator’s website.



Player screen: Welcome and waiting screen

Top of the screen:

- ❏ Current round indication and logout.
- ❏ In yellow, the table and player ID
- ❏ In gray, the house you will move into is empty until you choose where to live in the corresponding step.
- ❏ In blue, to track your game status at every time, you can always check the satisfaction and spendable income scores.
- ❏ **Satisfaction points:** Every player starts with an initial score of x satisfaction points. At the end of the game, satisfaction points are added for the remaining spendable income.
- ❏ **Spendable Income:** It is calculated every round but starts with the initial savings randomly assigned according to the player ID and welfare type.

Middle of the screen:

- ❏ Welcome and waiting messages.

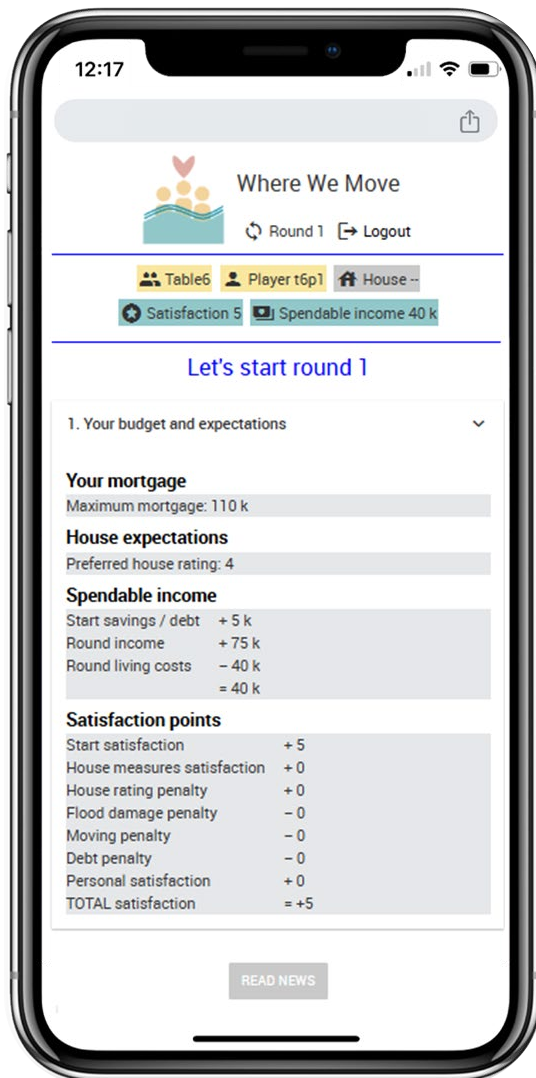
Bottom of the screen:

- ❏ Start game button is gray until the facilitator START NEW ROUND.

Seven steps every round

Step 1 - Player's budget and expectations

Explain on the player's first screen that each game round has seven steps. Describe the components of the budget screen, noting that the values marked with "*" remain the same throughout all rounds.



Player screen: Budget and expectations

Your mortgage

- ❏ **Max Mortgage***: Players can borrow up to this amount for a house. Anything above, players pay with the spendable income or become debt.
- ❏ **House mortgage**: House value paid, if any yet.
- ❏ **Left mortgage**: The total money players should still pay back to the bank after buying their house. 10% mortgage payments are automatically deducted each round.

House expectations

- ❏ **Preferred house rating***: House quality goal whose gap with the purchased house affects satisfaction once; players can buy improvements, so they shouldn't take on debt to reach it.

Satisfaction costs

- ❏ **Satisfaction increase per point***: What players pay can get in satisfaction for the remaining spendable income at the end of the game.

Spendable income

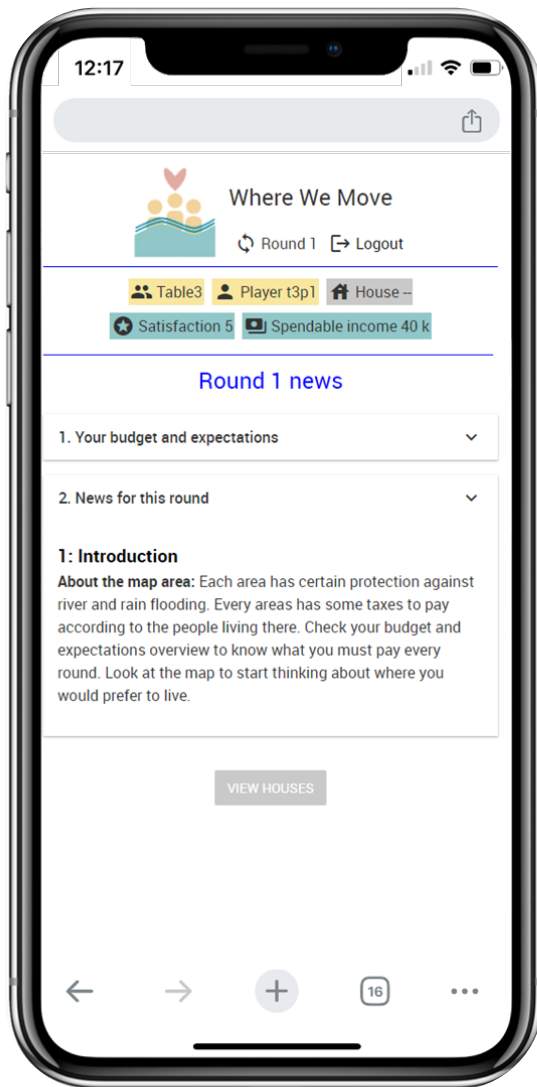
- ❏ **Starting savings**: It can become a debt (negative) according to your game choices.
- ❏ **Round income***: Received every round.
- ❏ **Round living costs***: Paid every round. Represents personal and daily house expenses (food, house services, etc.).

Satisfaction points

- ❏ **Start satisfaction**: At the beginning of each round.
- ❏ **House measures satisfaction**: Depending on the flood related improvements made at that step.
- ❏ **Penalties**: Satisfaction drops when players incur debt, take damage, move, or live on a lower house rating.
- ❏ **Personal satisfaction**: Changes from now on according to the only-satisfaction choices!

Step 2 – Round news

Every round starts with the news that changes the game situation, following either the municipal mandate or the overall trends at the place you move in. **ANNOUNCE NEWS** by clicking the button on step 2 on the game facilitators' website so that players can click on the **READ NEWS** button in their screen.



Player screen: News for this round

- ❏ Let the players read their news and clarify if any questions.
- ❏ In short, the news every round is as the table shows:

Round	Short news description
0	Welcome message
1	Move into an affordable house and increase your satisfaction
2	Climate change lowers protection levels per community; flood insurance becomes available.
3	Tax, satisfaction and flood protection increase in Naturcity
4	Taxes Naturcity +5k (add taxes and flood increase for Naturcity)

Step 3a – House market: Show houses, allow and finish buying

Open the house market by clicking the **SHOW HOUSES** button in **step 3** on the game facilitators' website:



Housing Game Facilitator App

Game session: Test
Group / Table: Table2
Round: 1
State: ANNOUNCE_NEWS

1. Rounds

2. News

3. House market

4. House improvements

5. Player perceptions

6. Floods and Damage

7. Summary

Show the available houses to the players on the table; press the **SHOW HOUSES** button for players to explore options. Go to your **HOUSE OVERVIEW** tab.

In the map, ensure every player puts the pawn on the preferred house. Do a bidding process if needed.

When everyone has agreed on the house to buy, press the buy button for every player. When finished, press the **ASSIGN HOUSES** and **CALCULATE TAXES** buttons.

SHOW HOUSES
ALLOW SELLING
FINISH SELLING
ALLOW BUYING
FINISH BUYING
CALCULATE TAXES

PLAYER OVERVIEW

HOUSE OVERVIEW

FLOOD OVERVIEW

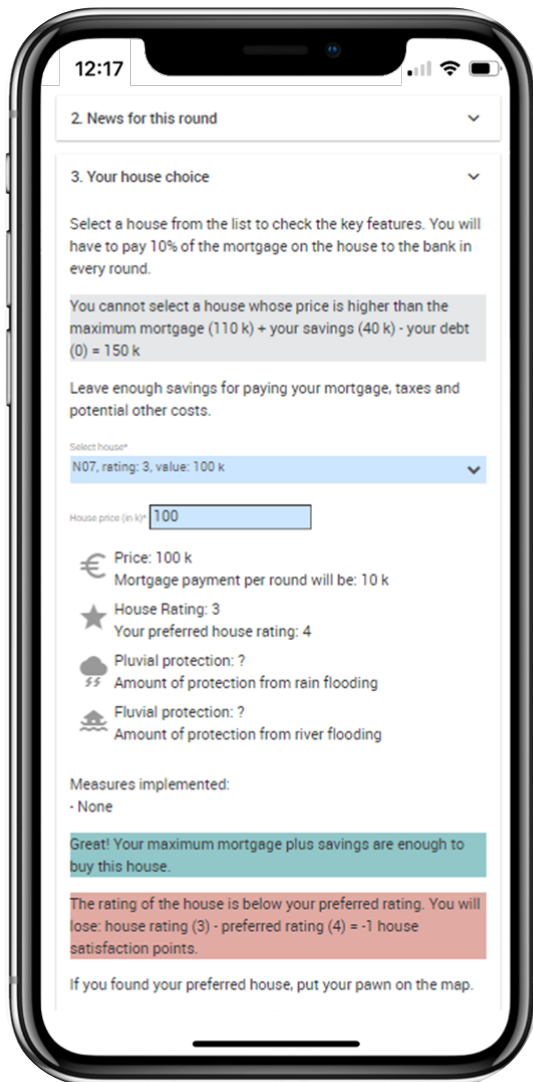
NEWS OVERVIEW

LOGOUT

Available houses in this round

Address	Available round	Rating	Market price	Buy price	House satisf.	Measures	Owner
N01	1	9	425 k	--	0	--	--
N04	1	6	200 k	--	0	--	--
N07	1	3	100 k	--	0	--	--
D01	1	8	300 k	--	0	--	--
D04	1	5	160 k	--	0	--	--
D05	1	5	160 k	--	0	--	--
D10	1	2	80 k	--	0	--	--
U01	1	6	200 k	--	0	--	--
U04	1	4	125 k	--	0	--	--
U10	1	2	70 k	--	0	--	--

Facilitator screen: House market



Player screen: View houses

Your house choices:

- ❏ **Reminder in gray highlight:** Even though they can buy a house up to their maximum mortgage plus savings, they should keep the money to pay the community taxes (see map) and the house mortgage (10% of the house value); otherwise, they will go into debt earlier or later in the round!
- ❏ **The blue drop-down menu:** Shows the houses available for this round as listed on the game facilitators' website.
- ❏ **The dark blue highlight:** Indicates otherwise in red if maximum mortgage house is enough (or not) to buy the house chosen.
- ❏ **The light red highlight:** Indicates otherwise in red if the house satisfaction rating can be achieved with the housing choice.

To keep in mind:

- ❏ **Facilitators can only approve a non-affordable house,** if the player has no other option.
- ❏ **Ask players to put down their pawns simultaneously on the board!** To do so, let the players think about which house they want for about 1-2 minutes

If two or more players select the same house, they bid against each other by asking for a bid card to fill and showing their bid only to the bank (facilitator).

PlayerID:	House:
Bid:	

Example bid card

- ❏ **Each value is allowed.** The bank sells at the market value - flood discount, if any is applicable.
- ❏ **The highest bid wins.** If players bid the same, they can bid again. Players can reject their bid house but must pick a different available house, and the rejected one goes to the next highest bidder.
- ❏ **If a player has insufficient Mortgage + Spendable income to buy the house,** the remainder will become a debt. A 1-point satisfaction penalty will apply as many rounds as the debt is negative. It is only possible if no other option.

When all players have agreed on the house to buy, click the ALLOW BUYING button on the facilitator's website below.

- ❑ If a bid was made, the player can adjust the house price accordingly as agreed upon in the players' website screen 'View houses'.
- ❑ On the facilitator's website, the facilitator can approve or deny the request as necessary after checking the player's request.



Housing Game Facilitator App
 Game session: Test
 Group / Table: Table2
 Round: 1
 State: ALLOW_BUYING

1. Rounds
 2. News
 3. House market
 4. House improvements
 5. Player perceptions
 6. Floods and Damage
 7. Summary

Show the available houses to the players on the table; press the SHOW HOUSES button for players to explore options. Go to your HOUSE OVERVIEW tab.

In the map, ensure every player puts the pawn on the preferred house. Do a bidding process if needed.

When everyone has agreed on the house to buy, press the buy button for every player. When finished, press the ASSIGN HOUSES and CALCULATE TAXES buttons.

SHOW HOUSES
 ALLOW SELLING
 FINISH SELLING
 ALLOW BUYING
 FINISH BUYING
 CALCULATE TAXES

PLAYER OVERVIEW
HOUSE OVERVIEW
 FLOOD OVERVIEW
 NEWS OVERVIEW
 LOGOUT

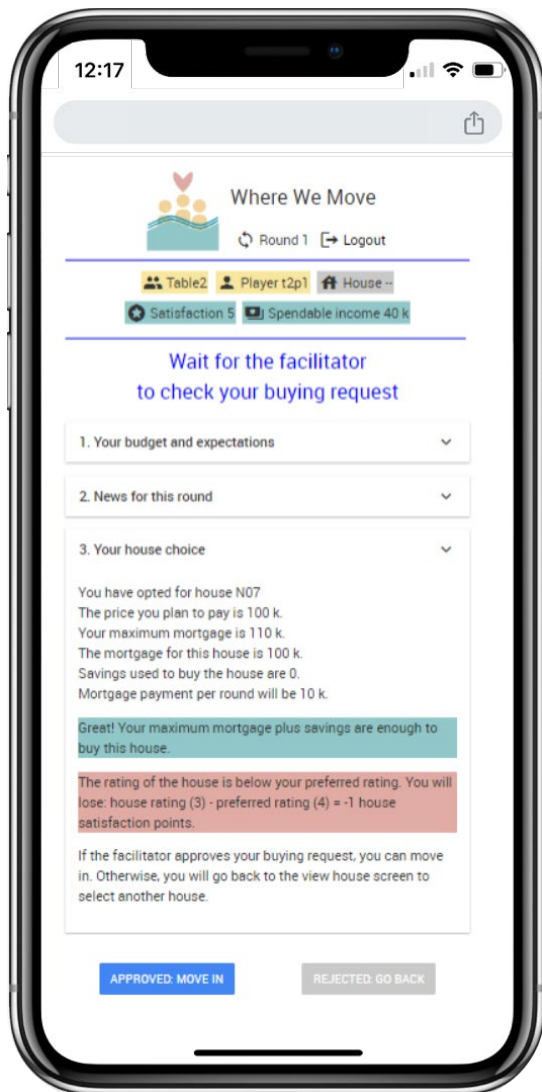
Available houses in this round

Address	Available round	Rating	Market price	Buy price	House satisf.	Measures	Owner
N01	1	9	425 k	--	0	--	--
N04	1	6	200 k	--	0	--	--
N07	1	3	100 k	--	0	--	--
D01	1	8	300 k	--	0	--	--
D04	1	5	160 k	--	0	--	--
D05	1	5	160 k	--	0	--	--
D10	1	2	80 k	--	0	--	--
U01	1	6	200 k	--	0	--	--
U04	1	4	125 k	--	0	--	--
U10	1	2	70 k	--	0	--	--

Buying requests
 Ensure all players have sufficient time to send their requests before handling them.
 This way, requests that need the facilitator's attention can be flagged as a note in red.

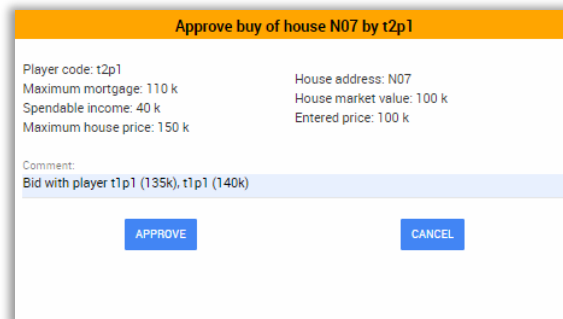
Player	Maximum mortgage	Savings / Debt	Maximum price	Selected house	Market price	Player's buy /bid price	Note	Approval	Rejection
t2p1	110 k	40 k	150 k	N07	100 k	100 k	OK	APPROVE	REJECT

Facilitator screen: Buying requests and selling



Player screen: Your house choice to click according to the facilitator's request reply.

- ❏ Facilitators can reject the request and redirect the player back to the view and buy houses screen when an error is made or the player changes their mind before the facilitator finishes the buying.
- ❏ The player's website shows accordingly a button to move in if the request is approved or to go back if the request is denied.
- ❏ Check that every player has bought the house (sent a house-buying request) and click the move into the house button on their player's screen, after you approve the request in the facilitator's screen.
- ❏ Facilitators can reject the request and redirect the player back to the view and buy houses screen when an error is made or the player changes their mind before the facilitator finishes the buying.
- ❏ The player's website shows accordingly a button to move in if the request is approved or to go back if the request is denied.
- ❏ Check that every player has bought the house (sent a house-buying request) and click the move into the house button on their player's screen, after you approve the request in the facilitator's screen.



Facilitator screen: Approving a house pop-up.

Step 3b – House market: Allow selling and buying

- Starting in round 2, an extra screen lets players decide whether to relocate by selling their house. Players first select the house to buy before offering to sell.
- Moving as flood adaptation measure is only encouraged in the last round. In the last round houses with protection measures become available for players to buy.
- Moving takes time, so players must state their motivation. If players relocate to a different house, players will lose 1 satisfaction, then a drop-down menu ask players to choose their reason for moving.
- When the coordinator activates the flood option, a price discount applies. Players can then sell only to the bank at the discounted value. For instance, with a 10k discount on a 100k house, the sell/bid price stays 100k. This option is inactive by default.
- After buying a house, players see a confirmation screen with their choices, payments, and satisfaction changes.

12:17

Your options for a new house

You can sell your house or stay in your current house. Moving will cost you a one-time moving penalty. To be able to move, you need to find an affordable house. You will pay 10% of the house price as the round mortgage.

You cannot select a house whose price is higher than the maximum mortgage (110 k) + your savings (43 k) + profit when selling your house (8 k est.) = 161 k

Note that your savings depend on the exact sales price of your house (the above estimate is based on the sales value according to the bank). Also leave enough savings for paying your mortgage, taxes and potential damage after flooding.

Select house*

Selling price

Based on the house's affordability options, do you still want to move? Then, you should sell your current house by agreeing on the selling price with the facilitator (if you sell to the bank) or with another player.

Selling price (in k) *

80

Once you set the price, wait for the facilitator to check and approve it!

Reason for selling *

Reason (in case of "Other")

STAY SELL HOUSE

Player screen: your options for a new house and selling price

12:17

Where We Move

Round 1 Logout

Table2 Player t2p1 House N07

Satisfaction 4 Spendable income 30 k

[View the status of your house](#)

1. Your budget and expectations

2. News for this round

3. Your house confirmation

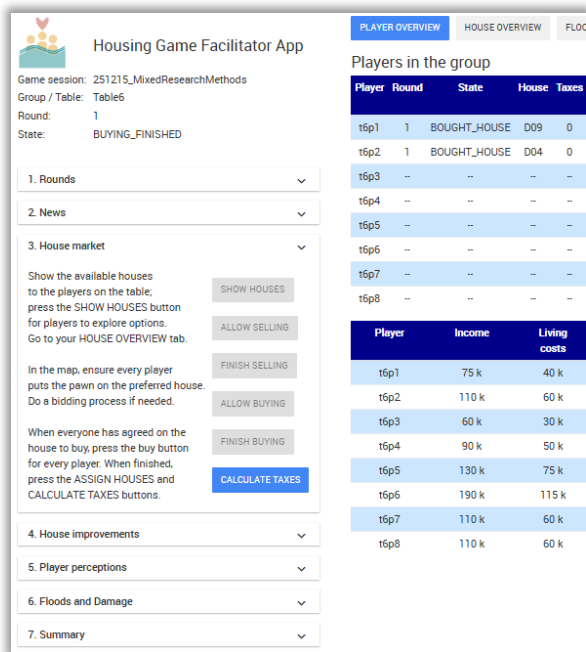
You live in house N07
You have bought a house in this round.
The price you paid is 100 k.
The left mortgage is 90 k.
Your maximum mortgage is 110 k.
Savings used to buy the house are 0.
Your preferred house rating is 4.
The rating of the house is 3.
Satisfaction change: -1 points.

VIEW TAXES

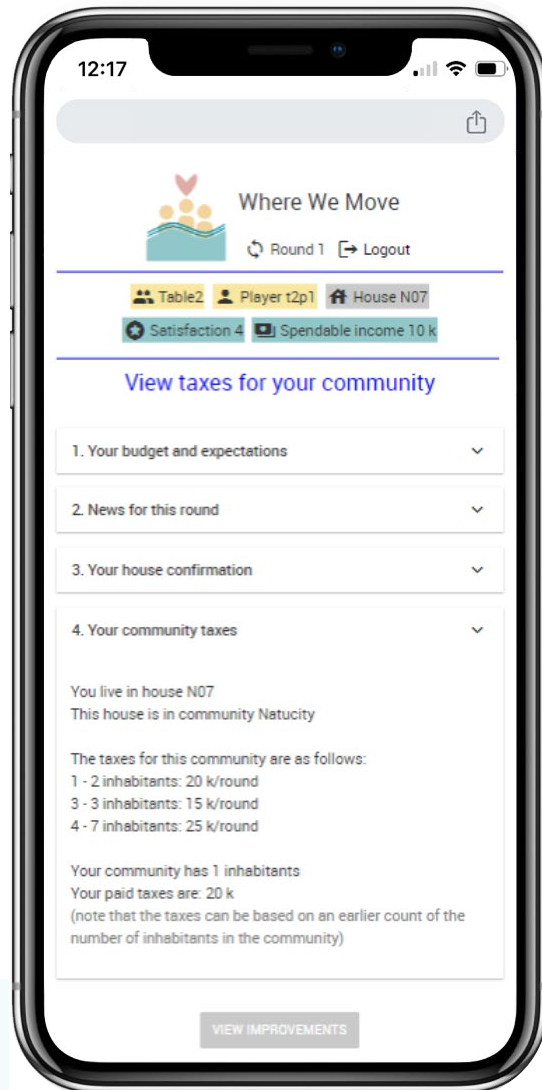
Player screen: your house confirmation with a summary of the new house type

Step 3c – Calculate taxes to show step 4: Community taxes for players

- ❏ Every active player has moved into a house. Check on the facilitator website have a “Bought house status” to help accordingly the player behind to finish with the house market.
- ❏ Click on the CALCULATE TAXES button according to the house where every player lives.
- ❏ The Community taxes player screen shows the applicable tax payments per number of inhabitants in the players community.
- ❏ Ask players to check the amount that they paid for taxes so that they can check how much their available income is after paying their taxes.



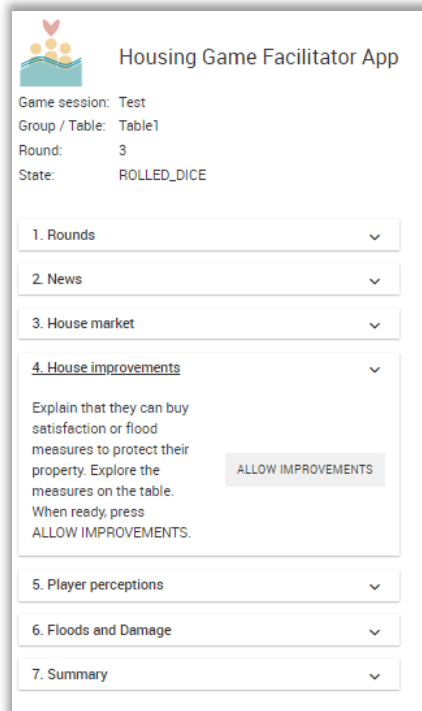
Facilitator screen: All active players' house choices.



Player screen: your community taxes with a summary of what players paid according to their community choice.

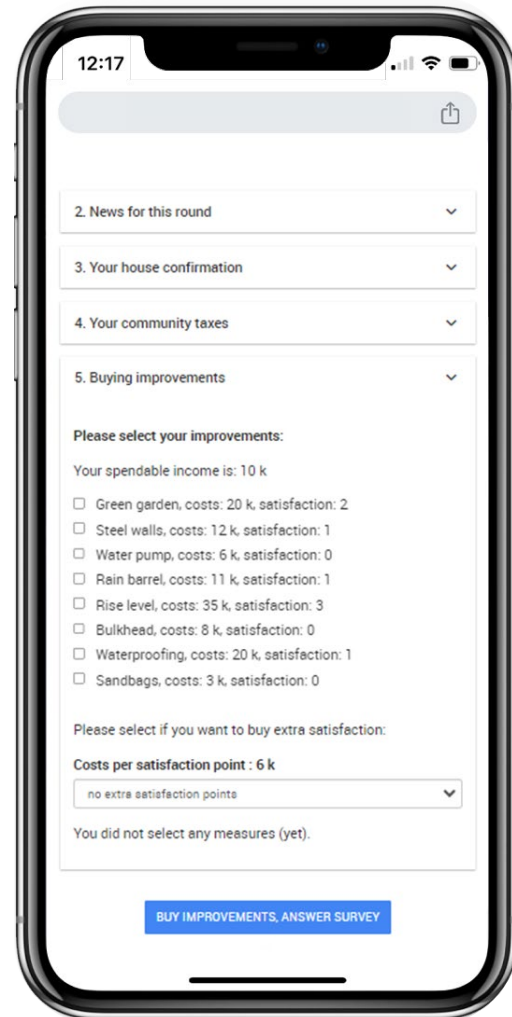
Step 4 – Allow improvements to show step 5: Buy improvements for players

- ❏ Click on the facilitators' website on the ALLOW improvements and ask players to decide on what to do with their spendable income according to the available choices.
- ❏ Remind the players that improvements stay with the house unless these are personal. So, if the player relocates, protection and satisfaction from the improvements made are lost from that house.



The screenshot shows the 'Housing Game Facilitator App' interface. At the top, it displays game session details: 'Game session: Test', 'Group / Table: Table1', 'Round: 3', and 'State: ROLLED_DICE'. Below this is a vertical list of menu items: '1. Rounds', '2. News', '3. House market', '4. House improvements', '5. Player perceptions', '6. Floods and Damage', and '7. Summary'. The '4. House improvements' item is selected and expanded, showing instructions: 'Explain that they can buy satisfaction or flood measures to protect their property. Explore the measures on the table. When ready, press ALLOW IMPROVEMENTS.' To the right of these instructions is a grey button labeled 'ALLOW IMPROVEMENTS'.

Facilitator screen: Allow improvements

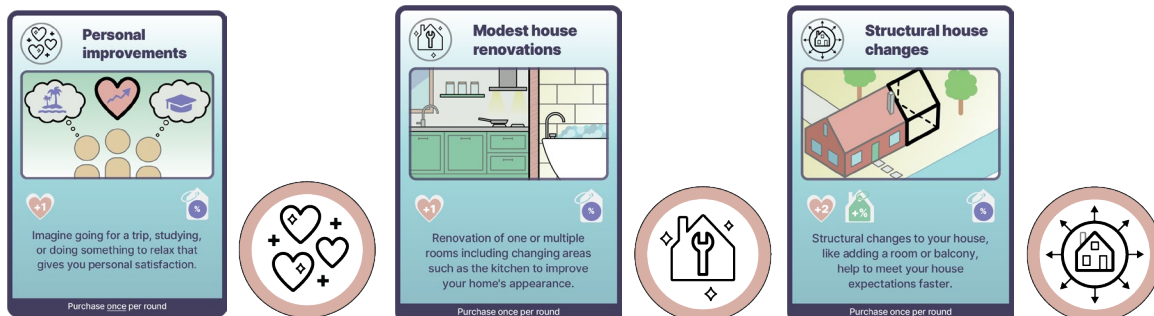


The screenshot shows a mobile phone screen displaying the player's interface. At the top, the time is 12:17. Below the status bar is a list of menu items: '2. News for this round', '3. Your house confirmation', '4. Your community taxes', and '5. Buying improvements'. The '5. Buying improvements' item is selected and expanded, showing a section titled 'Please select your improvements:'. Below this, it states 'Your spendable income is: 10 k'. There is a list of improvement options, each with a checkbox, name, cost, and satisfaction value: 'Green garden, costs: 20 k, satisfaction: 2', 'Steel walls, costs: 12 k, satisfaction: 1', 'Water pump, costs: 6 k, satisfaction: 0', 'Rain barrel, costs: 11 k, satisfaction: 1', 'Rise level, costs: 35 k, satisfaction: 3', 'Bulkhead, costs: 8 k, satisfaction: 0', 'Waterproofing, costs: 20 k, satisfaction: 1', and 'Sandbags, costs: 3 k, satisfaction: 0'. Below this list is another section titled 'Please select if you want to buy extra satisfaction:'. It states 'Costs per satisfaction point : 6 k' and has a dropdown menu currently set to 'no extra satisfaction points'. At the bottom, it says 'You did not select any measures (yet)'. A blue button at the very bottom is labeled 'BUY IMPROVEMENTS, ANSWER SURVEY'.

Player screen: Buying improvements

According to the improvement cards information, measures are permanent, available until used such as sandbags or should be bought every round such as the flood insurance. Ask players to place or remove as necessary the measures coin bought in their house card.

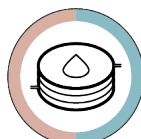
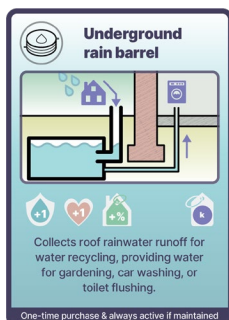
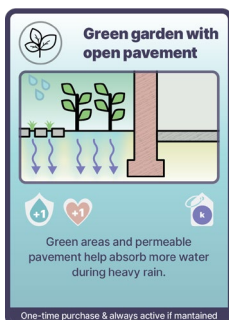
- ❖ **Flood is not a priority for every player.** Players may consider to improve their satisfaction by doing small or larger renovations or simply doing personal improvements that makes them happy.



- ❖ **Spending on flood adaptation:** The players can buy flooding measures to protect against rain and river flooding events later this round. The protection level is in addition to the public flood protection already available in the community where the player lives.
- ❖ **Flood adaptation measures can either prevent the water to enter before or during the flood to reduce damage.**



- ❑ Protection measures that waterproof walls and floors or increase the water storage give some satisfaction.



- ❑ Flood insurance is an option to consider to reduce the damage costs that should be bought every round to remain active:



Step 5 – Collect player' risk perception by showing step 6: Answer survey to players

Whether or not the facilitator clicks on **SHOW SURVEY**, the questions appear automatically once they finish making improvements. However, responses are collected after the facilitator clicks on **COMPLETE SURVEY**. Ask players to choose among the two options available to reply about:

Q1: about house protection

How do you judge the flooding risk of your house after the improvements you made?

- a. I won't get flooded
- b. If I get flooded, I won't get damaged
- c. I might suffer minor damage
- d. I will suffer some damage
- e. I will get seriously damaged

Q2: about confidence in community protection

How confident are you that the flood protection from the public measures in your community will protect your house against flooding?

- a. Absolutely not confident
- b. Not confident
- c. Inconclusive
- d. Somehow confident
- e. Absolutely confident

The screenshot shows the 'Housing Game Facilitator App' interface. At the top, there's a logo with three people icons. Below it, the app name is displayed. The session details are: Game session: Test, Group / Table: Table2, Round: 1, State: ALLOW_IMPROVEMENTS. A list of menu items is shown with dropdown arrows: 1. Rounds, 2. News, 3. House market, 4. House improvements, 5. Player perceptions (highlighted), 6. Floods and Damage, and 7. Summary. Under '5. Player perceptions', there's a text box that says 'Invite all players to enter the answers to the survey questions. Wait till all players have completed before closing'. To the right of this text are two buttons: 'SHOW SURVEY' (blue) and 'COMPLETE SURVEY' (grey).

Facilitator screen (top): Collect player perceptions

Player screen (right): Answer survey questions

The screenshot shows a mobile phone screen displaying the 'Answer survey questions' interface. At the top, the time is 12:17. Below the status bar, there's a header with player information: Table2, Player t2p1, House N07, Satisfaction 4, and Spendable income 4 k. The title 'Answer survey questions' is in blue. Below it, there's a list of survey questions with dropdown arrows: 1. Your budget and expectations, 2. News for this round, 3. Your house confirmation, 4. Your community taxes, 5. Your bought improvements, and 6. Answer survey (highlighted). Under '6. Answer survey', there's a text box that says 'Please answer the following questions:'. Below this, there are two questions: 'Question 1. How do you judge the flooding risk of your house?' with a dropdown menu showing '2 I won't get damaged', and 'Question 2. Do you trust that the public measures in your community will protect your house against flooding?' with a dropdown menu showing '2 No'. At the bottom, there's a blue button labeled 'STORE SURVEY'.

Step 6a – Flood event

This step determines whether a river/rain flood happens. Hence, the facilitator should check on its screen that all active players have completed the survey before rolling the dices.

The screenshot shows the 'Housing Game Facilitator App' interface. On the left, a sidebar menu lists seven sections: 1. Rounds, 2. News, 3. House market, 4. House improvements, 5. Player perceptions, 6. Floods and Damage (selected), and 7. Summary. The main content area is divided into two panels. The top panel, titled 'Players in the group', shows a table with columns 'Player', 'Round', and 'State'. The bottom panel, titled 'Floods and Damage', contains instructions for rolling dice and two input fields for 'River dice roll (1-12)' and 'Rain dice roll (1-10)', with a 'ROLL DICE, SHOW DAMAGE' button. The right sidebar shows two tabs: 'PLAYER OVERVIEW' (selected) and 'HOUSE OVERVIEW'. The 'PLAYER OVERVIEW' tab displays a table of player income.

Player	Round	State
t6p1	2	SURVEY_COMPLETED
t6p2	2	SURVEY_COMPLETED
t6p3	--	--
t6p4	--	--
t6p5	--	--
t6p6	--	--
t6p7	--	--
t6p8	--	--

Player	Income
t6p1	75 k
t6p2	110 k
t6p3	60 k
t6p4	90 k
t6p5	130 k
t6p6	190 k
t6p7	110 k
t6p8	110 k

River flood:

- ❏ To determine whether a river flood will occur in one of the communities, two six-sided dice are rolled by the players. The dice choice is because river floods are less frequent and more damaging.

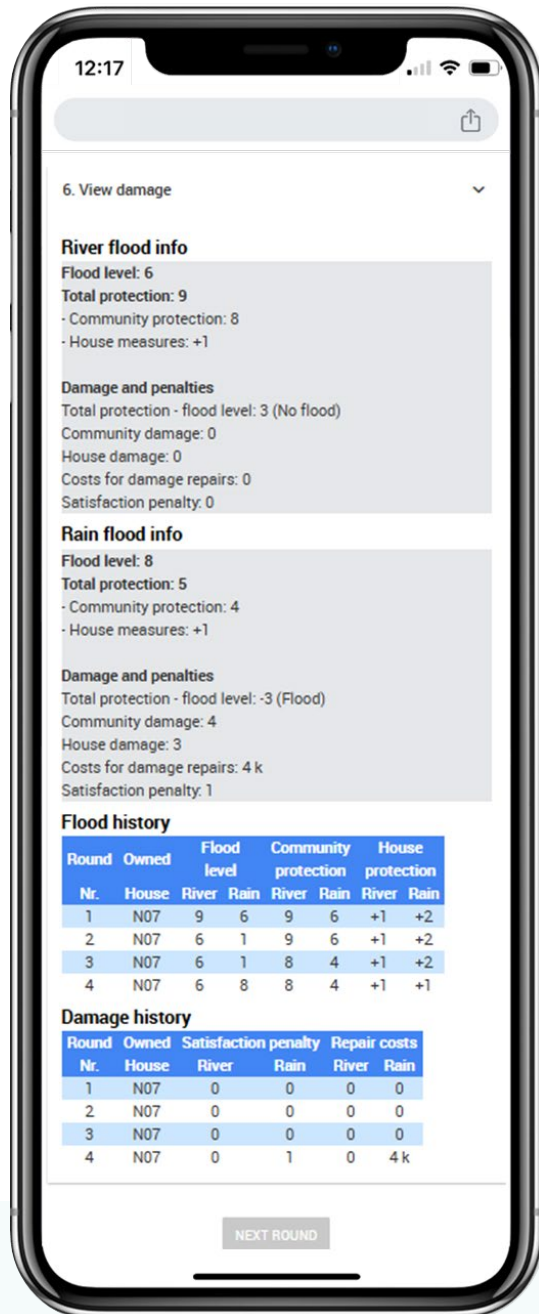
Rain flood:

- ❏ Unlike river floods, rain floods are less impactful but are more common. Hence, we use a 10-sided dice.

Facilitator screen: Roll dice

Step 6b – View damage

Now explain how the game determines each round if a River and Rain flooding happens and how private measures can help protect a player's house.



Player screen: View damage

Whether of flood or rain, damage calculations are as follows:

- ❏ **Total protection:** (Community protection + house measures) – dice value: water volume players are unprotected for.
- ❏ **House damage points:** Flood damage happens only when (protection – dice roll) is negative.

River flooding:

- ❏ **Repair costs** equal the number of damage points multiplied by the repair cost per point (4k).
- ❏ **Satisfaction penalty:** Damage points + 1 if the community is flooded.

Rain flooding:

- ❏ **Repair costs** are always equal to one repair cost point (4k) if there is a rain flood to account for the less impactful damage.
- ❏ **Satisfaction penalty** is always equal to 1 if there is a rain flood. There is no extra satisfaction penalty if the area is flooded.

Flood history:

- ❏ Players can see the flood levels, community and house protection every round.

Damage history:

- ❏ Lists per round satisfaction penalty and repair costs.

The Flood Overview tab in the facilitator's website allows to check the flood protection level per house community and house.

Dice roll thrown every round.

Per-round public protection in each community and applicable house discount when enabled.

Display the per-round details in the table below for every house available in the game.

Housing Game Facilitator App

Game session: 251007_VerzekeraarsMasterClass
 Group / Table: Table1
 Round: 4
 State: SHOW_SUMMARY

1. Rounds
 2. News
 3. House market
 4. House improvements
 5. Player perceptions
 6. Floods and Damage
 7. Summary

PLAYER OVERVIEW HOUSE OVERVIEW **FLOOD OVERVIEW** NEWS OVERVIEW LOGOUT

Dice roll			Naturcity			Dyketown			Unbesvillage		
Round nr.	River	Rain	Round nr.	House discount	Public protection	House discount	Public protection	House discount	Public protection	View details	
1	9	6	1	–	River: 9, Rain: 6	–	River: 9, Rain: 4	–	River: 6, Rain: 6	Details	
2	6	1	2	–	River: 9, Rain: 6	–	River: 9, Rain: 4	–	River: 6, Rain: 6	Details	
3	6	1	3	–	River: 8, Rain: 4	–	River: 8, Rain: 2	–	River: 5, Rain: 4	Details	
4	6	8	4	–	River: 8, Rain: 4	–	River: 8, Rain: 2	–	River: 5, Rain: 4	Details	

Selected data for round 4

House code	From round	Owner	Measures	Base protection		Public delta (news)		Private delta (measures)		Damage points	
				River	Rain	River	Rain	River	Rain	River	Rain
N01	1	–		9	6	-1	-2	0	0	0	4
N04	1	t1p2	Flood insurance (R4) Green garden (R1) Waterproof walls, floors (R2) Sandbags (R3) Underground rainbarrel (R3)	9	6	-1	-2	3	4	0	0
N05	1	–	Self-activating wall (R1) Green garden (R2) Underground rainbarrel (R2) Waterproof walls, floors (R2)	9	6	-1	-2	2	4	0	0
N07	1	t1p1	Green garden (R1) Sandbags (R4)	9	6	-1	-2	1	1	0	3

Facilitator screen: Flood overview

Lists for the selected round the house protection indicating the measures implemented and the damage points used as basis for the player screen. repair costs and satisfaction penalty.

Step 7 – View summary

Every round, the spendable income is re-calculated according to the player's choices. Before moving to the next round players should **paying off their debt, including any damages they had.**

Spendable income

Start savings / debt	+ 1 k
Round income	+ 75 k
Round living costs	- 40 k
Actual mortgage payment	- 10 k
Actual taxes	- 20 k
Personal improvements	- 5 k
House improvements	- 1 k
House damage	- 4 k
	= -3 k

Spendable income:

- ❏ Start savings or debt:
 - +/- Spendable income previous round
 - +/- Profit or debt for buying or selling a house
- ❏ Round Income received every round
- ❏ Calculated Costs:
 - Living costs paid every round
 - Actual mortgage for the house moved in
 - Actual taxes for the house moving in
 - Round Mortgage paid
 - Spent savings/debt for buying house
 - Only satisfaction improvements
 - House improvements due to flood adaptation
 - House damage due to river and rain floods

Every round, the satisfaction points are re-calculated according to the player's choices:

Satisfaction points

Start satisfaction	+ 13
House measures satisfaction	+ 0
House rating penalty	+ 0
Community living bonus	+ 2
Flood damage penalty	- 1
Moving penalty	- 0
Debt penalty	- 0
Personal satisfaction	+ 2
TOTAL satisfaction	= +16

Satisfaction points:

- ❏ +/- Satisfaction previous round
- ❏ Satisfaction improvements
 - + House (flood adaptation) measures
 - + Personal (only satisfaction) measures
- ❏ Satisfaction penalties =
 - Flood damage due to river and rain floods
 - Debt
 - Moving
- +/- House rating difference only applicable when moving in

Game Settings Summary

Game news text in English and Dutch

The table below shows the news the facilitator reads.

	Facilitator's text English	Facilitator's text Dutch
Round 0 Welcome message	Where We Move is a housing game to take the homeowners' perspective when joining efforts for climate action. The goal is to maximize satisfaction through your choices or leftover income. As you play, you'll discover which decisions best protect your home and yourself within the set budget and expectations.	Where We Move is een woonspel waarin je vanuit het perspectief van huiseigenaren samenwerkt aan klimaatactie. Het doel is om de tevredenheid te maximaliseren via je keuzes of overgebleven inkomen. Tijdens het spel ontdek je welke beslissingen je huis en jezelf het beste beschermen binnen het beschikbare budget en de gestelde verwachtingen.
Round 1 Move into an affordable house and increase your satisfaction	Where do we move on the game map? There are three communities. Diketown and Unbesvillage each has 10 houses. Naturcity has only 7 houses because it has more nature. Not all houses are available to buy now. What is the flood protection on each area? Naturcity offers good protection for river flood and is the best in the game against rain floods as nature areas help to drain rainwater. Diketown offers also good protection for rivers but the lowest for rain as the more houses are built the less space to drain rain. Unbesvillage offers similar protection to Naturcity against rain floods but has the lowest for river floods as it is next to the water channels without additional protection. What is your budget and expectations? Every round, you receive income and pay living costs. Keep about 50k in savings for the round mortgage, taxes, improvements, and unpredictable damage. Otherwise, you may pay a debt penalty.	Waar bewegen we ons op de spelkaart? Er zijn drie gemeenschappen. Diketown en Unbesvillage hebben elk 10 huizen. Naturcity heeft slechts 7 huizen omdat het meer natuur bevat. Niet alle huizen zijn op dit moment beschikbaar om te kopen. Bescherming tegen overstromingen: Naturcity biedt goede bescherming tegen rivieroverstromingen en is de beste in het spel tegen regenoverstromingen. Diketown biedt ook goede bescherming tegen rivieroverstromingen, maar heeft de laagste bescherming tegen regen. Unbesvillage biedt vergelijkbare bescherming als Naturcity tegen regenoverstromingen, maar heeft de laagste bescherming tegen rivieroverstromingen. Wat is jouw budget en wat zijn je verwachtingen? Elke ronde ontvang je inkomen en betaal je woonkosten. Houd ongeveer 50k aan spaargeld achter de hand voor de hypotheek van de ronde, belastingen, verbeteringen en onvoorspelbare schade. Anders kun je een schuldenboete krijgen.

	Facilitator's text English	Facilitator's text Dutch
Round 2 Flood damages explanation reminder	This is a new round, so you should remove any flood insurance from your house card and take away any sandbags you may have used. Any questions on how to calculate the flood damage?	Dit is een nieuwe ronde, dus je moet eventuele overstromingsverzekering van je huiskaart verwijderen en de zandzakken die je hebt gebruikt weghalen. Vragen over hoe je de overstromingsschade berekent?
Round 3 Climate change lowers protection levels per community; flood insurance becomes available.	More rain on the way: More heavy rainfall over the next years will result in more floodings. Climate change lowers protection levels of all communities by -1 for the river and -2 for the rain protection.	Meer regen op komst: De komende jaren zal zwaardere neerslag leiden tot meer overstromingen. Door klimaatverandering daalt de beschermingsgraad van alle gemeenschappen met -1 voor rivierbescherming en -2 voor regenbescherming.
Round 4 Taxes Naturcity +5k (add taxes and flood increase for Naturcity)	Naturcity taxes rise by 5 for the municipality to fund area upgrades so that nature allows for water storage and boosts water quality, cuts air pollution, and protects habitats—benefiting people and biodiversity. Residents get a one-time 2-point satisfaction bonus and gain 1 point when moving there (instead of losing 1). Ask the facilitator to take out the house cards that will become available this round, some of which have already measures	De Naturcity-belasting stijgt met 5 punten zodat de gemeente gebiedsverbeteringen kan financieren, waardoor natuur meer water kan bergen en de waterkwaliteit verbetert, luchtvervuiling vermindert en habitats beschermt—tot voordeel van mens en biodiversiteit. Bewoners ontvangen een eenmalige tevredenheidsbonus van 2 punten en krijgen 1 punt bij verhuizing naar dit gebied (in plaats van 1 punt verlies). Vraag de facilitator om de huiskaarten uit te nemen die deze ronde beschikbaar komen; sommige hiervan hebben al maatregelen.

House market characteristics

The tables below show an overview of all available houses in the game. Houses with measures are only available in the last round. All other houses are available in all other rounds but encouraged to buy only in the first round.

Naturcity

House code	House price	House rating	Initial river flood protection	Initial rain flood protection	Active measures
N01	425k	9	0	0	
N02	200k	6	1	1	
N03	425k	9	1	1	
N04	200k	6	0	0	
N05	200k	6	0	0	
N06	100k	3	1	0	
N07	100k	3	0	0	

Diketown

House code	House price	House rating	Initial river flood protection	Initial rain flood protection	Active measures
D01	300k	8	0	0	
D02	300k	8	0	0	
D03	300k	8	0	0	
D04	160k	5	0	0	
D05	160k	5	0	0	
D06	160k	5	2	2	 
D07	160k	5	2	2	 
D08	80k	2	1	2	 
D09	80k	2	0	0	
D10	80k	2	0	0	

Unbesvillage

House code	House price	House rating	Initial river flood protection	Initial rain flood protection	Active measures
U01	200k	6	0	0	
U02	200k	6	1	1	
U03	200k	6	1	2	 
U04	125k	4	0	0	
U05	125k	4	0	0	
U06	125k	4	1	0	
U07	125k	4	2	2	 
U08	70k	2	1	1	
U09	70k	2	0	0	
U10	70k	2	0	0	

Game Credits

This game has been developed by the TUDelft GameLab thanks to the funding from the NWO project “Crossing the Borders at the Grensmaas”, the 4TU.DeSIRE Resilience Program, the TU Delft Climate Action Program and the TUDelft Digital Competence Centre. The following interdisciplinary team of collaborators have so far contributed accordingly.

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